Zhuoyun Du

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1 RESEARCH INTERESTS

My research asks whether agents can **learn to design their own communication protocols, coordination rules, and improvement mechanisms**, rather than relying on human-specified ones. I pursue this through three connected threads:

1. **Latent-Space Communication for Agentic Foundation Models:** Treating latent representations as a native medium for reasoning and coordination among agents, and as a bridge between agent design and core representation learning.
2. **Self-Improving Multi-Agent Systems:** Building agents that refine coordination and problem-solving through interaction, feedback, and textual parameter optimization, moving beyond fixed workflows.
3. **Autonomous Evaluation & Post-Training:** Exploring agent-driven evaluation, preference alignment, and data-generation loops that let models improve in open-ended settings with less dense human supervision.

2 EDUCATION

Zhejiang University, Hangzhou, China — Sep 2024 -- Jul 2027 (Expected)
M.Eng. in Data Science and Engineering. — GPA: 89.5 / 100 (Overall Ranking: 4 / 50).
State Key Lab of CAD & CG. Advisor: Prof. Wei Chen.

Jinan University, Guangzhou, China — Sep 2020 -- Jul 2024
B.Eng. in Computer Science and Technology. — GPA: 90 / 100; (Rank: 3 / 99).

3 SELECTED PUBLICATIONS

† indicates co-first / equal contribution.

First-author / co-first-author

1. **Zhuoyun Du**†, Runze Wang†, Huiyu Bai, Zouying Cao, Xiaoyong Zhu, Bo Zheng, Wei Chen and Haochao Ying. Enabling Agents to Communicate Entirely in Latent Space [**Interlat**]. ***ACL 2026***. [paper] [Code]
2. **Zhuoyun Du**†, Lujie Zheng†, Renjun Hu, Yuyang Xu, Xiawei Li, Ying Sun, Wei Chen, Jian Wu, Haolei Cai, and Haochao Ying. LLMs can simulate standardized patients via agent coevolution [**EvoPatient**]. ***ACL 2025***. [paper] [Code]
3. **Zhuoyun Du**†, Chen Qian†, Wei Liu, Zihao Xie, Yifei Wang, Yufan Dang, Weize Chen, and Cheng Yang. Multi-agent software development through cross-team collaboration [**Croto**]. ***Findings of ACL 2025***. [paper] [Code]
4. Huiyu Bai†, Runze Wang†, **Zhuoyun Du**†, Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng and Xuejiao Zhao. Online-PVLM: Advancing Personalized VLMs with Online Concept Learning [**Online-PVLM**]. ***Submitted to MM 2026***. [arXiv]

Co-authored

1. Shan He, Runze Wang, **Zhuoyun Du**, Huiyu Bai, Zouying Cao, Yu Cheng, Bo Zheng. Learning to Evolve: A Self-Improving Framework for Multi-Agent Systems via Textual Parameter Graph Optimization [**Learning to Evolve**]. ***Findings of ACL 2026***. [paper]
2. Yuewen Liu, Peng Xu, **Zhuoyun Du**, Muxi Diao, Anyi Zhang, Yang Li, Yutong Zhang. MemCoRL: Alternating Co-Optimization of Memory Retrieval and Utilization via Collaborative Reinforcement Learning [**MemCoRL**]. ***ACL 2026***.
3. Chen Qian†, Zihao Xie†, Yifei Wang†, Wei Liu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, Zhiyuan Liu, and Maosong Sun. Scaling large-language-model-based multi-agent collaboration [**MACNet**]. ***ICLR 2025***. [paper]
4. Wei Liu†, Chenxi Wang†, Yifei Wang, Zihao Xie, Rennai Qiu, Yufan Dang, **Zhuoyun Du**, Weize Chen, Cheng Yang, and Chen Qian. Autonomous agents for collaborative task under information asymmetry [**iAgent**]. ***NeurIPS 2024***. [paper]

4 RESEARCH & INDUSTRY EXPERIENCE

ByteDance, Data Department — Mar 2026 -- Present
Talent Program Intern.

- Built the team's **first** data-flywheel rollout system for multi-turn, task-oriented dialogue, supporting downstream **multi-turn dialogue RL** and **agentic RL**.

- Designed a **self-play simulation and evaluation environment** with high-fidelity, persona-conditioned user-simulation agents for rollout generation and evaluation signals.
- Conducted research on **LLM post-training and preference alignment** (SFT, RLHF/DPO) and autonomous agent-driven evaluation of dialogue policies.

Alibaba, Future Living Lab / Token Foundry with Tongyi Jan 2025 -- Mar 2026
Research Intern.

- Worked in a foundation-model ecosystem whose broader initiatives include HappyHorse (#1 on Artificial Analysis Video Arena) and HappyOyster (real-time interactive world model), focusing on latent-space communication for multi-agent reasoning.
- Proposed **Interlat**, replacing natural-language message passing with temporally aligned hidden-state exchange; designed token-latent curriculum and learned compression, achieving up to 24× speedup and gaining hands-on 256+ GPU distributed training experience.
- Co-developed **Online-PVLM** ***Submitted to MM 2026***; co-authored **MemCoRL** ***ACL 2026*** and **Learning to Evolve** ***Findings of ACL 2026***.

Zhejiang University, State Key Lab of CAD & CG Sep 2024 -- Present
M.Eng. Research, advised by Prof. Wei Chen.

- Led **EvoPatient** ***ACL 2025***, an open-source doctor-patient coevolution system for virtual standardized patients, piloted in clinical teaching and OSCE-style training at hospitals including the Second Affiliated Hospital of Zhejiang University School of Medicine.
- Drove research formulation, hospital-side requirement gathering, coevolution-method design, implementation, prompt engineering, experiments, medical-case curation/de-identification, expert evaluation, and paper writing.
- Improved patient-answer *Ability* score from 0.763 (Few-shot baseline) to 0.860 after 200 training cases. [paper]

Tsinghua University, THUNLP Dec 2023 -- Aug 2024
Research Assistant.

- Conducted research within the **OpenBMB/ChatDev** ecosystem (**33k+ GitHub stars**) under Prof. **Zhiyuan Liu**, studying role organization, information exchange, and coordination among LLM agents.
- Led **Croto** ***Findings of ACL 2025*** as a ChatDev-branch project, designing cross-team orchestration with parallel teams, hierarchy partitioning, and greedy aggregation.
- Contributed to **MACNet** ***ICLR 2025*** and **iAgent** ***NeurIPS 2024***, covering scalable collaboration and information-asymmetry settings.

SKILLS, HONORS, SERVICE & BACKGROUND

- **LLM & Machine Learning:** Multi-agent system design; distributed training on **256+ GPUs** (*e.g.,* A100, A800, H100, H20); SFT, LoRA, RLHF/DPO; DeepSpeed, FlashAttention.
- **Programming Languages & Frameworks:** Python, C/C++, Bash, Git, LaTeX; PyTorch, HuggingFace ecosystem (*e.g.,* Transformers, Datasets).
- **Languages:** English (IELTS: 7.5); Mandarin (Native).
- **Honors:**
  - 2025 **National Scholarship**
  - Outstanding Graduate Student Scholarship
  - **3** Undergraduate **Scholarships**
  - **First-Class Scholarship for Outstanding Graduates**
- **Service:** Program Committee, NeurIPS 2025 and ICLR 2025.
- **Interests & Background:** Experienced in collaborative research and communication; former **National Champion in Amateur Épée Fencing**; Grade 7 Piano Certificate; interests include snowboarding, photography, and arts and culture.